

canon-0.8.1-research-track

Companion research-track sectors migrated from the main SST canon

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Abstract

This companion document collects those sectors that remain useful for continuity and future development but are too long, too provisional, or too phenomenology-specific for the core body of *Swirl-String-Theory_Canon-v0.8.1*. Their migration out of the main canon is editorial rather than negative: each subsection remains part of the broader SST research program, but none is required for the minimal logical closure of the core canon.

Editorial note (v0.8.1): the maximal swirl tension sector ($F_{\text{swirl}}^{\text{max}}$, Rydberg bridge identities, Gibbons-class maximum-tension comparison, and the threefold interpretation table) is developed in the core document under `SST_CANON-v0.8.1.tex`, subsection `Maximal Swirl Tension` (label `sec:Fmax`); it is not duplicated here.

1 Research-track sectors migrated from the main canon

1.1 Thermodynamic Radiation Interface

A recurring legacy idea is that a local swirl-energy density may define an effective temperature field and thereby an emergent radiation sector. Let

$$T_{\text{eff}} = T_{\text{eff}}(\rho_E, \omega, S_{(t)}^{\circ}) \quad (1)$$

denote an unspecified constitutive relation. A minimal legacy interface then asks whether Planck-type spectra can emerge from a medium whose local excitation scale is set by T_{eff} .

[ORTHODOX]: blackbody spectra are governed by Planck's law once a valid temperature variable is defined.

[SPECULATIVE]: SST does not yet derive the constitutive relation above, so this sector remains a research program rather than canonically closed physics.

1.2 Minimal Coupling Interface to QED

Early research notes repeatedly explored whether an effective vector potential could be built from circulation variables. The most conservative placeholder takes the form

$$\nabla \rightarrow \nabla - i \frac{m}{\hbar} A_{\text{eff}}, \quad A_{\text{eff}} = \chi v, \quad (2)$$

where χ is a scalar weight and v is the local carrier velocity field.

[ORTHODOX]: minimal coupling is the standard gateway from free to gauge-coupled wave dynamics.

[SPECULATIVE]: the above substitution is only an analogy template unless a full gauge-covariant derivation is supplied.

1.3 Ribbon-Invariant Chirality Refinement

The older canon layers also suggested using the Calugareanu–White–Fuller relation

$$Lk = Tw + Wr \quad (3)$$

as a refinement of the knot-based matter/antimatter dictionary.

[ORTHODOX]: this is a standard geometric identity for framed curves and ribbons.

[DERIVED]: it can be used internally to separate writhe-dominated from twist-dominated realizations of the same coarse knot class.

[SPECULATIVE]: the mapping from $\text{sign}(Tw)$ or related chirality measures to particle/antiparticle sectors remains a model hypothesis.

1.4 Particle-Candidate Mapping Layer

[SPECULATIVE] Working dictionary: a minimal knot/composition mapping retained from earlier SST work is summarized in Table 1. This table is organizational rather than definitive; it records the current topological assignments used by the mass and taxonomy program.

Knot class / composition	Candidate	Status / role
3_1	electron	baseline lepton calibration
$T(2, 5)$ or equivalent higher torus class	muon	lepton hierarchy candidate
$T(2, 7)$ or equivalent higher torus class	tau	lepton hierarchy candidate
$5_2 + 5_2 + 6_1$	proton	composite baryon candidate
analogous nearby composite class	neutron	composite baryon candidate

Table 1: Minimal knot-class to particle-candidate summary retained as a research-track dictionary.

[CRITICAL NOTE]: Table 1 is a working dictionary only. A full canonical assignment requires an explicit invariant set and a mass-functional comparison program.

1.5 Hydrodynamic Exchange and the Pauli Barrier

[ORTHODOX]: for two thin filamentary loops C_1 and C_2 in an incompressible medium, the Biot–Savart interaction energy is

$$E_{\text{int}} = \frac{\rho_f \Gamma_1 \Gamma_2}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\ell_1 \cdot d\ell_2}{\|r_1 - r_2\|}. \quad (4)$$

[DERIVED]: if two identical electron-candidate loops are forced into the same spatial state, regularization at the core radius r_c yields the overlap penalty

$$V_{\text{Pauli}}(d) \approx \frac{\rho_f \Gamma_0^2}{4\pi} \iint \frac{d\ell_1 \cdot d\ell_2}{\sqrt{\|r_1(\theta_1) - r_2(\theta_2) + d\|^2 + r_c^2}}, \quad (5)$$

with maximal-overlap scale

$$V_{\text{Pauli}}^{\text{max}} \approx \frac{\rho_f \Gamma_0^2}{4\pi} \left(\frac{L}{r_c} \right) \mathcal{O}(1). \quad (6)$$

Using the canonical values

$$\rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}, \quad (7)$$

$$r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad (8)$$

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_\odot\| \approx 9.6836 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}, \quad (9)$$

and $L \sim 2\pi a_0$ at the hydrogenic scale, one obtains the benchmark estimate

$$V_{\text{Pauli}}^{\text{max}} \sim 1.23 \times 10^{-18} \text{ J} \sim 7.6 \text{ eV}. \quad (10)$$

[CRITICAL NOTE]: this is retained as a canonical scale benchmark, not yet as a closed theorem. Its role is to show that the exclusion-energy estimate falls in an atomically relevant window rather than at obviously pathological scales.

1.6 Numerical Benchmark Layer

[DERIVED] Computational-program requirement: earlier SST work contained explicit benchmark tables for particle and atomic masses. In v0.8.1, the benchmark layer is retained as a reproducibility requirement rather than as a stand-alone authority claim.

Particle	m_{exp} (MeV)	m_{SST} (MeV)	rel. error	mode class
e	0.51099895	0.51099895	0	predictive/closure
μ	105.6583745	...	...	predictive candidate
τ	1776.86	...	...	predictive candidate
p	938.272088	...	...	predictive or closure
n	939.565420	...	...	predictive or closure

Table 2: Benchmark summary structure to be populated from canonical scripts and archived CSV output.

[CRITICAL NOTE]: a future v0.8.x benchmark appendix should include the exact script or package path used for generation, archived CSV outputs, mode definitions (predictive vs. exact-closure), and uncertainty propagation on canonical constants and geometric factors.

1.7 Helicity-by-base archive (diagnostic layer)

[DERIVED] Archive role: the SST helicity-by-base outputs are retained as a measurable balance diagnostic for taxonomy support, not as an independent law. Define the midpoint-distance diagnostic

$$\Delta_H(K) = \left| \widehat{\mathcal{H}}_b(K) + \frac{1}{2} \right|, \quad (11)$$

where $\widehat{\mathcal{H}}_b(K)$ is the archived normalized helicity-like base indicator for class K .

Near-balanced values with $\Delta_H(K) \ll 1$ indicate candidate dark or quasi-dark sectors only when symmetry and instability filters are also satisfied.

[CRITICAL NOTE]: no single helicity-like scalar is sufficient for sector assignment; classifier decisions remain conjunction rules over symmetry, helicity balance, and low-order stability content.

[CANON WORKFLOW NOTE]: if SST-labelled and parallel VAM-labelled helicity outputs are numerically identical, the SST-labelled archive is the only required source for canon workflows.

1.8 Trefoil-closure archive (benchmark discipline)

[DERIVED] Benchmark role: ideal-trefoil robustness sweeps, final-best-estimate tables, and theorem-style reports are retained as benchmark-discipline infrastructure.

[CRITICAL NOTE]: practical best-estimate closure and theorem-level closure are distinct. A result may remain useful for calibration while still failing strict theorem-level tail or postcheck requirements.

[REQUIRED METADATA]: benchmark reports should include explicit pass/fail logic, postchecks, extrapolation summaries, and local-curve diagnostics so that closure claims are auditable.

1.9 ϕ_{3_1} transform archive (deferred)

[SPECULATIVE] Status: retained as research-track spectral evidence for later zeta/spectral programs.

[SCOPE RULE]: this archive is not part of the current dark-sector or galactic canonization closure path in v0.8.x.

1.10 Gauge-Sector Roadmap

[SPECULATIVE] Structural roadmap: a retained minimal statement of the gauge program is

$$g_{\text{eff}} = su(3) \oplus su(2) \oplus u(1), \quad (12)$$

with effective gauge phases organized as holonomy data of multi-director transport.

A minimal phase-based representation is

$$A_\mu^a(x) = \partial_\mu \theta^a(x), \quad (13)$$

and clock coupling acts as a common phase reference,

$$\theta^a(x) \mapsto \theta^a(x) + q^a \chi(x) \quad \Rightarrow \quad A_\mu^a \mapsto A_\mu^a + q^a \partial_\mu \chi. \quad (14)$$

[CRITICAL NOTE]: the gauge layer is retained only as a roadmap statement. Running couplings, anomaly cancellation, and realistic mixing matrices remain open work.

2 Dark-sector and galactic canonization execution package (v0.8.x)

2.1 Stage-0 freeze: topology toolkit and benchmark protocol

[DERIVED] Scope freeze: the mandatory topology toolkit for v0.8.x canonization is fixed to:

1. Gauss linking integral,
2. helicity in ideal fluids,
3. thin-tube helicity reduction,
4. Hopf invariant,
5. basic torus-knot invariants (c, b, g) ,
6. Calugareanu–White–Fuller relation $Lk = Tw + Wr$.

[DERIVED] Notation freeze: one notation family is used in this package:

$$\rho_f, \quad v_\theta(r), \quad p_{\text{swirl}}(r), \quad \mathcal{H}(K), \quad N_u^{(1)}(K), \quad \Gamma, \quad \kappa, \quad S_{(t)}^\circ.$$

No parallel symbol aliases are introduced for the same quantity.

[DERIVED] Mandatory benchmark protocol template:

1. **Inputs:** canonical constants, topology identifiers, profile parameters, and dataset/version tags.

2. **Procedure:** exact script or package path, run command, and branch/commit hash.
3. **Outputs:** archived CSV files with fixed column schema and an attached metadata note.
4. **Uncertainty:** first-order propagation for calibrated constants and sensitivity sweep on profile parameters.
5. **Rerun:** deterministic rerun steps and acceptance tolerances.

[CANON WORKFLOW NOTE]: every promoted quantitative claim in this package is invalid without an explicit input-to-CSV trace.

2.2 Stage-1 freeze: dark-sector Euler pressure anchor

[ORTHODOX] Euler radial balance (steady, axisymmetric, azimuthal-dominant):

$$\frac{1}{\rho_f} \frac{dp_{\text{swirl}}}{dr} = \frac{v_\theta(r)^2}{r}. \quad (15)$$

[DERIVED] Flat-tail limit (constant tail speed):

$$v_\theta(r) \rightarrow v_0 \implies p_{\text{swirl}}(r) = p_0 + \rho_f v_0^2 \ln\left(\frac{r}{r_0}\right). \quad (16)$$

[DERIVED] Assumption block: incompressible, inviscid, stationary, axisymmetric, azimuthal-dominant flow, with validity restricted to regions where radial drift and stratification corrections are subleading.

[DERIVED] Dimensional check:

$$\left[\frac{1}{\rho_f} \frac{dp}{dr} \right] = \frac{\text{kg m}^{-2} \text{s}^{-2}}{\text{kg m}^{-3}} = \text{m s}^{-2} = \left[\frac{v^2}{r} \right].$$

[SPECULATIVE] SST interpretation: Eq. (15) is interpreted as a classifier/driver layer for dark-sector support. This interpretation is model-level and separate from the orthodox derivation.

[DERIVED] Attached observables/falsifiers:

1. Rotation-curve consistency after pressure reconstruction from archived $v_\theta(r)$ data.
2. Null falsifier: systematic mismatch in the reconstructed pressure gradient sign or scale across validated radial windows.

2.3 Stage-2 freeze: measurable dark taxonomy

[DERIVED] Minimal classifier variables: symmetry class (including amphichirality status), chirality sign/state, helicity-balance indicator $\Delta_H(K)$, and low-order instability count $N_u^{(1)}(K)$.

[DERIVED] Decision rules (symmetry-first, then stability-filtered):

$$\text{Dark} : \text{amphichiral candidate} \wedge \Delta_H(K) \ll 1 \wedge N_u^{(1)}(K) = 0, \quad (17)$$

$$\text{Quasi-dark} : \text{near-balanced symmetry/helicity} \wedge 0 < N_u^{(1)}(K) \leq 2, \quad (18)$$

$$\text{Visible} : \text{chiral-dominant and/or helicity-unbalanced sectors}. \quad (19)$$

[DERIVED] Ambiguity tie-break: if classes overlap, choose the class with the strongest conjunction score in the order

symmetry gate $\rightarrow$ helicity balance gate $\rightarrow$ stability gate.

[DERIVED] Baseline templates and falsifiers:

- **Visible baseline:** trefoil-like chiral template; falsifier = persistent amphichiral-plus-balanced signature under validated data updates.
- **Quasi-dark baseline:** Borromean-type weak-coupling template; falsifier = stable zero-instability assignment across independent scans.
- **Dark baseline:** figure-eight/amphichiral template; falsifier = robust nonzero chirality or high-instability assignment.

[CANON WORKFLOW NOTE]: amphichiral $\Rightarrow$ dark remains a classifier proposal tested against observables, not a closed theorem.

2.4 Stage-3 freeze: galactic law branch gate and closure route

[DERIVED] Branch-gate decision: Branch A (legacy profile refine) is selected for current v0.8.x closure because it preserves tested limits while allowing benchmark traceability.

Working profile:

$$v_\theta(r) = \frac{C_{\text{core}}}{\sqrt{1 + (r_c/r)^2}} + C_{\text{tail}} \left(1 - e^{-r/r_c}\right). \quad (20)$$

[DERIVED] Required checks: small- r behavior, flat-tail behavior, parameter semantics, and pressure reconstruction via Eq. (15).

[DERIVED] Failure modes:

1. non-physical parameter combinations (wrong sign or unstable growth),
2. failure to satisfy tail-limit behavior in benchmark windows,
3. inability to reconstruct a consistent pressure profile from the same archived inputs.

2.5 Stage-4 freeze: probe falsification map

[DERIVED] probe classes are fixed as astrophysical, laboratory pressure/swirl analog, HV-coil/thrust-inspired, optical/torsional coupling, and null dark-sector tests.

[CANON WORKFLOW NOTE]: probe outcomes can support or reject already-stated equations, but they do not replace derivation-level closure.

2.6 Promotion snapshot after execution

[DERIVED] post-freeze status in this package:

- Item 6 (topology toolkit): CANON-ready in scope and notation.
- Item 5 (benchmark protocol): CANON-CANDIDATE pending repository-wide protocol reuse.
- Item 1 (Euler pressure anchor): CANON-CANDIDATE with explicit assumptions, limits, and falsifier path.
- Item 2 (dark taxonomy): BRIDGE with measurable classifier gates and falsifiers.
- Item 3 (galactic swirl law): BRIDGE with branch gate closed and benchmark route fixed.
- Item 4 (probe map): CANON-BRIDGE with equation/null-result mapping frozen.

Probe class	Equation test	under profile	Null-result meaning	Primary con-founder	Expected sign/trend
Astrophysical rotation curves	Eq. (15) + closure		weakens pressure-law applicability window	baryonic-model bias	reconstructed dp/dr consistent with v_θ^2/r
Lab swirl analog	Euler radial balance scaling	bal-	weakens scale-transfer claim	viscosity/edge forcing	monotone pressure rise with imposed swirl
HV-coil/thrust-inspired	circulation-pressure claims	bridge	kills quantized-bridge claim if no scaling	EM/thermal artifacts	response tracks circulation proxy
Optical/torsional coupling	swirl-clock signatures	bridge	weakens coupling interpretation	detector drift/noise	phase/timing shift with controlled swirl proxy
Null dark tests	taxonomy conjunction rules	con-	kills classifier branch if conjunction fails	mislabelled topology class	class score decreases under contradictory diagnostics

Table 3: Frozen equation-to-probe map for falsification-first workflow in the v0.8.x dark-sector program.